

Supplementary Material for SCCM: Spherically Consistent Coarse Matching for ERP Dense Feature Correspondence

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A Implementation Details

Reproducibility. Tab. 1 summarizes the shared training protocol used for all models retrained on Matterport3D [2] (the Holo360D training protocol is in Sec. P); SCCM changes only the coarse-matcher modules, leaving the loss, encoder, and refiner architecture unchanged.

Numerical TPB details. For a ray $\mathbf{r} = (\cos \varphi \sin \lambda, \sin \varphi, \cos \varphi \cos \lambda)$, the orthonormal tangent frame $\mathbf{B}_{\mathbf{r}} = [\mathbf{e}_{\varphi} \ \mathbf{e}_{\lambda}]$ (Sec. 4.1, main) is

$$\mathbf{e}_{\varphi} = \frac{\partial \mathbf{r}}{\partial \varphi} = (-\sin \varphi \sin \lambda, \cos \varphi, -\sin \varphi \cos \lambda), \quad \mathbf{e}_{\lambda} = \mathbf{e}_{\varphi} \times \mathbf{r}. \quad (1)$$

$\mathbf{e}_{\varphi}$ has unit norm for every (φ, λ) , including the poles; $\mathbf{e}_{\lambda}$ is automatically unit and orthogonal since $\mathbf{r} \perp \mathbf{e}_{\varphi}$, and is aligned with the $+\lambda$ direction. We read $(\sin \lambda, \cos \lambda)$ off the ray as $x/\cos \varphi, z/\cos \varphi$ with $\cos \varphi = \sqrt{\max(1 - y^2, \epsilon_b)}$ ($\epsilon_b = 10^{-7}$, distinct from the LAC clamp $\epsilon = 10^{-3}$ in Eq. (5), main); these ratios stay finite because $x, z \rightarrow 0$ together at the poles, so the frame is well-defined at every coarse-grid location (grid token centers never sample the exact poles, so this clamp is effectively inactive). We then evaluate the spherical log-map (Eq. (3), main) in a stable form with an inner-product clamp (keeping $\theta < \pi$) and a norm clamp (removing the 0/0 pole form), so the resulting bias is finite for every token pair. Because TPB forms an all-to-all ($N \times N$) table on the 32×64 coarse grid, exact antipodal pairs occur; these are only $O(N)$ of the $O(N^2)$ entries, and the norm clamp assigns them a bounded, deterministic bias value, so $\delta_{q,k}$ is geometrically exact away from this sparse set.

Encoding and MLP. Before Fourier encoding, the offset is scaled per query by $\cos^{\alpha} \varphi_q$ with a single learnable scalar α initialized at 0, which lets the bias adapt its latitude conditioning. The scaled offset is encoded with 8 geometric frequencies $(1, 2, \dots, 2^7)$ per tangent coordinate as $(\sin, \cos)$ pairs (32 dimensions) and mapped by a $32 \rightarrow 32 \rightarrow 1$ MLP (GELU, zero-initialized output; 1,089 parameters); α and this MLP are the only learned parts of TPB.

Table 1: Shared training protocol for all models retrained on Matterport3D, used for the fixed-scaffold ablations.

Setting	Value
Optimizer	AdamW ($\beta_1=0.9$, $\beta_2=0.999$, wd 0.01); non-zero LR only on trainable modules
LR schedule	base 1×10^{-4} , constant, $\times 0.1$ at step 112,500 (90% of the 125K steps)
Budget	1 M samples (≈ 125 K steps); all rows identical
Loss	RoMa [5] dense regression + certainty BCE + coarse-grid CE, unmodified (SCCM adds none)
GT corresp.	spherical warp from per-pixel depth + relative pose (intrinsic-free ERP)
MP3D split	official 61/11/18 scenes = 54,015/4,625/15,682 pairs; native 1024×512
Augmentation	pair-shared horizontal flip ($p = 0.5$) and independent per-frame yaw $\theta_A, \theta_B \in [0, 2\pi)$ (EDM [6] convention); no color/crop
Encoder	DINOv2-Large (patch 14), frozen; $448 \times 896 \rightarrow 32 \times 64$ tokens ($N=2048$)
Resolution	inputs resized to 448×896 ; metrics on the dense 448×896 refiner warp
Hardware	8xRTX 4090 (24 GB), batch 1/GPU, bf16 forward; ≈ 22 h/run

B Stanford2D3D Test Protocol

Frame source. Stanford2D3D [1] provides 1,413 equirectangular panoramas at 4096×2048 across 6 indoor areas (area 5 is split into 5a/5b as separately scanned wings). We drop one truncated panorama in area 3, leaving **1,412** usable panoramas (area 1: 190, area 2: 299, area 3: 84, area 4: 258, area 5a: 143, area 5b: 230, area 6: 208).

Coordinate harmonization. The dataset stores camera poses as 3×4 world-to-camera matrices in a $+y$ -down image-frame convention and depth as 16-bit PNGs with scale $1/512$ m. Two normalizations match our runtime ERP convention (Sec. 3, main): (i) a $\text{diag}(1, -1, 1)$ y-flip on each pose’s rotation block, so image-top corresponds to $+y$ (up); (ii) the $/512$ depth scaling to meters.

Pair generation. Within each area we enumerate all uni-directional pairs (i, j) , $i < j$, and compute an EDM-style overlap score:

$$\text{ov}(i, j) = \frac{|\{p \in V_i : |d_{i \rightarrow j}(p) - d_j(\Pi(p))| / d_j(\Pi(p)) < 0.1\}|}{|V_i|},$$

where V_i is the set of valid-depth pixels of I_i , Π is the depth-and-pose-driven ERP reprojection from I_i to I_j , and the relative-depth consistency threshold 0.1 follows EDM [6]. We retain pairs with $\text{ov}(i, j) \in [0.30, 0.80]$.

Resulting pair count.

Area	Frames	All pairs ($i < j$)	Retained ($0.30 \leq \text{ov} \leq 0.80$)
area 1	190	17,955	1,057
area 2	299	44,551	3,403
area 3	84	3,486	233
area 4	258	33,153	1,178
area 5a	143	10,153	620
area 5b	230	26,335	1,118
area 6	208	21,528	1,135
Total	1,412	157,161	8,744

All 8,744 pairs are used for evaluation.

Difference from EDM’s protocol. EDM [6] retains pairs with $\text{ov} > 0.50$ (3,460 pairs). Our $[0.30, 0.80]$ overlap band includes lower-covisibility pairs and excludes near-duplicate views.

C Downstream Evaluation

The main paper summarizes both downstream tasks (relative pose and 3D reconstruction) in Sec. 5.4 under one unified protocol. This section provides the full protocol, the overlap-selected pose subset (Tab. 2), and top-down reconstructions (Fig. 1).

Pose. For each Matterport3D test pair we sample 2,000 matches by certainty-weighted multinomial sampling (RoMa [5]), unproject them to unit-ray pairs, and estimate the essential matrix with the same 8-point + RANSAC solver for every method, decomposing it to $(\mathbf{R}, \mathbf{t})$ by cheirality; the pose AUC of Tab. 3 (main) is aggregated over the 11,574-pair subset defined under Reconstruction below. To control for pair difficulty, we additionally report an overlap-selected subset ($\text{ov} > 0.5$, 9,268 pairs; Tab. 2) under the same pipeline. All methods use this single evaluator, so differences reflect the matches rather than the estimator.

Reconstruction. For each test pair we triangulate stride-4 pixels of image A against their predicted matches under the GT pose (two-ray intersection, cheirality-filtered), using a shared per-pixel certainty gate $\tau = 0.5$ —the same certainty signal the pose pipeline samples from (RoMa [5], DKM [3]). All methods are scored on the common 11,574-pair subset where every method yields ≥ 200 confident matches; the triangulated cloud is compared to the GT depth cloud for accuracy, completeness, and F-score at 5/10/20 cm. The ungated variant (no gate) is discussed in Sec. N.1.

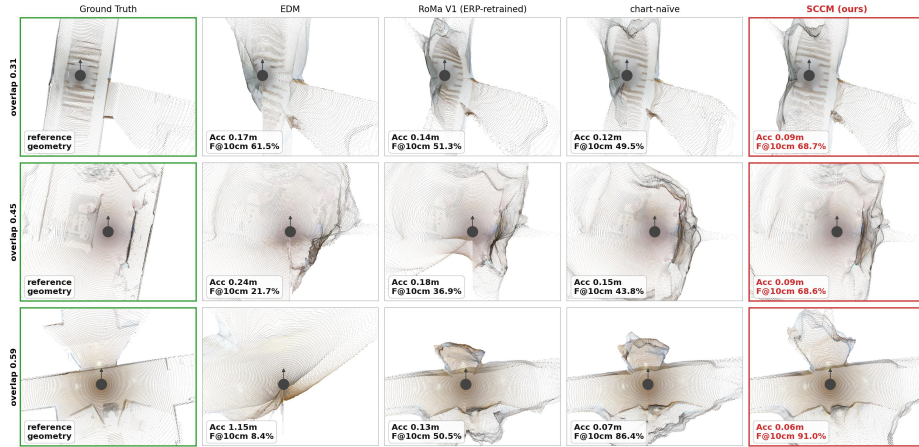


Fig. 1: Top-down reconstruction comparison on three representative test pairs (qualitative; aggregate reconstruction metrics are in Tab. 3, main). Each panel projects the triangulated cloud onto the X-Z plane (camera *A* at origin). On these examples the baselines (EDM, RoMa V1 (ERP-retrained)) and our chart-naïve scaffold produce distorted room contours, whereas SCCM’s cloud (red) aligns more closely with the GT reference (green).

Table 2: Pose on the overlap-selected subset ($ov > 0.5$, 9,268 pairs) under our unified ray-essential evaluator. We report Pose-AUC using $\max(\text{rotation}, \text{translation})$ error. Best **bold**, second underlined.

Method	AUC@5°	AUC@10°	AUC@20°
EDM [6]	11.52	24.55	38.17
RoMa V1 (ERP-retrained) [5]	14.69	33.60	54.28
chart-naïve	<u>17.56</u>	<u>37.68</u>	<u>57.97</u>
SCCM	22.06	42.36	61.17

D Performance Gains over the Chart-Naïve Baseline

Tab. 3 summarizes the SCCM – chart-naïve differences on Matterport3D and zero-shot Stanford2D3D. SCCM improves all five metrics on both datasets, with $\text{PCK}@1^\circ$ gains of 4.5 pp and 5.0 pp, respectively. Training variability is assessed in Sec. R.

E Sphere-Area-Weighted Metric

We recompute every metric with per-pixel weights proportional to $\cos \varphi$ to account for sphere area. Table 4 shows that the chart-naïve $\rightarrow$ SCCM gain persists on every metric under area weighting, on both Matterport3D and zero-shot Stanford2D3D.

Table 3: Gains over the chart-naïve baseline on Matterport3D test (15,682 pairs) and zero-shot Stanford2D3D (8,744 pairs). PCK is fraction-of-pixels ($\uparrow$); MAE/median are in degrees ($\downarrow$). Differences are computed from the displayed values. Best in **bold**.

Set	Method	PCK@1° $\uparrow$	PCK@3° $\uparrow$	PCK@5° $\uparrow$	MAE $\downarrow$	Med $\downarrow$
MP3D	chart-naïve	0.230	0.609	0.769	5.68	2.23
	SCCM	0.275	0.665	0.806	5.36	1.88
	Δ	+0.045	+0.056	+0.037	−0.32	−0.35
Stanford2D3D	chart-naïve	0.179	0.475	0.627	18.86	3.26
	SCCM	0.229	0.564	0.697	17.09	2.45
	Δ	+0.050	+0.089	+0.070	−1.77	−0.81

Table 4: Sphere-area-weighted metrics: each pixel weighted by $\cos \varphi$ vs the pixel-uniform grid, on Matterport3D test and zero-shot Stanford2D3D. SCCM leads all baselines (EDM, ERP-retrained RoMa V1, the chart-naïve scaffold) on every metric under both pixel-uniform and sphere-area-weighted aggregation, on both datasets. The pixel-uniform columns reproduce the main-paper tables; the area-weighted columns re-weight the same test pixels by $\cos \varphi$.

		PCK@1° $\uparrow$	PCK@3° $\uparrow$	PCK@5° $\uparrow$	MAE $\downarrow$	Med $\downarrow$
<i>Matterport3D test</i>						
EDM	pixel-uniform	0.163	0.400	0.509	16.78	4.79
	area-weighted	0.186	0.419	0.518	17.16	4.55
RoMa V1 (ERP-retrained)	pixel-uniform	0.198	0.538	0.711	6.60	2.76
	area-weighted	0.238	0.565	0.716	6.84	2.44
chart-naïve	pixel-uniform	0.230	0.609	0.769	5.68	2.23
	area-weighted	0.264	0.622	0.764	5.98	2.06
SCCM	pixel-uniform	0.275	0.665	0.806	5.36	1.88
	area-weighted	0.303	0.667	0.795	5.74	1.78
<i>Zero-shot Stanford2D3D</i>						
EDM	pixel-uniform	0.104	0.262	0.355	35.70	10.69
	area-weighted	0.136	0.302	0.387	35.11	9.68
RoMa V1 (ERP-retrained)	pixel-uniform	0.167	0.452	0.628	17.80	3.45
	area-weighted	0.215	0.494	0.643	17.27	3.06
chart-naïve	pixel-uniform	0.179	0.475	0.627	18.86	3.26
	area-weighted	0.221	0.504	0.638	17.99	2.96
SCCM	pixel-uniform	0.229	0.564	0.697	17.09	2.45
	area-weighted	0.274	0.588	0.710	16.25	2.20

F Off-Gravity Behavior Beyond Upright ERP

SCCM assumes upright ERP. To probe this assumption we synthesize off-gravity captures from the test set: for a fixed pitch rotation R we resample both panoramas by R and conjugate the relative pose $T' = GTG^{-1}$ ($G = \text{blkdiag}(R, 1)$), which preserves the GT correspondences up to resampling artifacts while tipping the chart off the gravity horizon. Table 5 reports PCK@1° for SCCM, its

chart-naïve scaffold, EDM, and ERP-retrained RoMa V1 on the full test split under $10^\circ/20^\circ/30^\circ$ pitch. All ERP matchers degrade off-gravity; SCCM is not tilt-equivariant, but owing to its larger upright margin it retains the highest *absolute* PCK@ 1° at every tested tilt. A natively tilt-equivariant matcher remains future work.

Table 5: Off-gravity tilt robustness on Matterport3D test (15,682 pairs). A fixed SO(3) pitch is applied to both views and the relative pose is conjugated accordingly, so the ground truth remains consistent up to resampling artifacts. We report PCK@ 1° under increasing pitch. All ERP matchers degrade off-gravity; SCCM is not tilt-equivariant but retains the highest absolute accuracy at every tested angle. Best per column in **bold**.

Method	upright (resampled)	pitch 10°	pitch 20°	pitch 30°
EDM [6]	0.163	0.105	0.044	0.018
RoMa V1 (ERP-retrained) [5]	0.198	0.096	0.042	0.022
chart-naïve	0.230	0.099	0.043	0.023
SCCM (ours)	0.273	0.142	0.064	0.034

G Rationale for Backbone and Baseline Training Choices

Why not V2. We use RoMa V1’s public training recipe for the controlled 1 M-sample protocol. A comparable public recipe for RoMa V2 [4] was unavailable at the time of this study. Moreover, V2’s intertwined positional changes (normalized-grid RoPE, fixed frequency $\omega = 1$, multi-view Transformer with alternating attention) interact directly with the attention-side modules of SPA, so re-applying SPA on V2 would confound “what the V2 positional changes provide” with “what our sphere-aware prior provides.”

V1 is the appropriate scaffold for controlled ablation. On the fixed chart-naïve attention scaffold (cross-attention + dual-softmax, no PE) that replaces V1’s GP coarse stage, we add our yaw-periodic RoPE, tangent-plane bias, and AAC cumulatively; this is the scaffold used for the cumulative ablation (Tab. 2, main) and the factorial decomposition (Sec. M). V2 is retained as a zero-shot perspective baseline in Tab. 1 (main), where its released inference checkpoint suffices.

EDM baseline. EDM’s [6] public release provides inference code and a checkpoint trained on Matterport3D; the training code, loss, and data pipeline are not released. Retraining EDM under our 1 M-sample protocol would therefore require re-implementing its full training stack, with the same reproduction confound noted for V2 above. We instead evaluate EDM from its released in-domain checkpoint (trained by its authors on the same Matterport3D data under their own recipe and budget), and separately evaluate its input-side absolute spherical

PE as a mechanism-only control retrained under the identical protocol (Tab. 2, main; Sec. J).

Portability to other matchers. SPA and AAC attach to two generic interfaces rather than to RoMa-specific components: SPA to the pairwise attention logits of the coarse matcher, and AAC to its per-pixel pre-sigmoid covisibility logits. Our chart-naïve scaffold R1 combines cross-attention, a covisibility head, and dual-softmax, and the cumulative ablation (Tab. 2, main) validates the modules on this scaffold. Transfer to other attention-based matchers, including LoFTR [12], requires support for the positional rotations and additive pairwise bias used by SPA, and a per-pixel covisibility logit for AAC. GP-based coarse stages (DKM [3], original RoMa V1 [5]) can accommodate AAC through their covisibility logits, but SPA additionally requires an attention-based coarse stage, as in R1.

H Yaw-Periodic RoPE: Frequency Choice

Restricting the longitudinal RoPE frequencies to integers ($\omega_i^\lambda \in \mathbb{Z}$, Sec. 4.1 (i) main) is the only choice consistent with 2π -periodicity at the chart seam: any fractional ω^λ shifts the cross-seam relative phase by a non-multiple of 2π and breaks the cyclicity SPA enforces. At coarse-grid width $W = 64$, the longitudinal Nyquist limit is 32 cycles per 360° . Our maximum frequency $\omega_{\max} = d_h/4 = 16$ ($d_h = 64$) is half this limit; its 22.5° period spans four token spacings. The integer-frequency design outperforms the geometric-frequency control in Tab. 2 of the main paper. Only the longitudinal half of each head is restricted; the latitude half keeps the standard geometric schedule ($\omega_i^\varphi = \theta_0^{-2i/(d_h/2)}$, $\theta_0 = 10000$), so fine latitude discrimination is preserved as in the latitude subspace of standard RoPE.

I Chart-naïve Coarse Matcher (R1 Backbone)

The chart-naïve backbone (row R1) keeps RoMa V1’s [5] frozen DINOv2-Large [9] encoder and the ConvRefiner *architecture* (the refiner weights, like all non-encoder modules, are trained from scratch), and replaces *only* V1’s Gaussian-process coarse matcher with a standard attention + dual-softmax coarse stage. It carries *no* sphere-aware component—no yaw-periodic RoPE, tangent-plane bias, EDM abs. PE, or log-area correction—and every ablation row, the EDM abs. PE control, and SCCM is this same backbone augmented with one or more such terms, which is what makes the comparison controlled.

Coarse stage. Coarse features $\mathbf{x}_A, \mathbf{x}_B \in \mathbb{R}^{N \times d}$ ($d = 512$, $N = H \times W$ the coarse grid) pass through a cascade of $n_{\text{blocks}} = 4$ Cross $\rightarrow$ Self attention blocks (8 heads, each with a feed-forward sub-layer) using *no* positional encoding and *no* relative-position bias. Assignment is by dual-softmax

$$\mathbf{P} = \text{softmax}_{\text{row}}(\mathbf{S}/\tau) \odot \text{softmax}_{\text{col}}(\mathbf{S}/\tau), \quad \mathbf{S} = \mathbf{x}_A \mathbf{x}_B^\top, \quad \tau = 0.1, \quad (2)$$

and the soft correspondence (the expected target coordinate under the row-normalized $\mathbf{P}$) is passed to the RoMa V1 ConvRefiner.

Covisibility gate. A per-query head produces a covisibility logit $\ell_p = \text{MLP}(\mathbf{x}_p)$ (128-d hidden) mapped to a gate $\mu_p = \sigma(\ell_p)$ that down-weights non-covisible queries. In R1 this is the bare MLP; AAC (Sec. 4.2, main) adds the single log-area term on top of this same logit. All components except the frozen encoder are randomly initialized and trained from scratch under the protocol of Sec. A.

J EDM Positional Embedding: Re-implementation

Scope of this probe. The purpose here is to isolate a single design axis—an *absolute, input-side* sphere-coordinate positional encoding (as in EDM) vs. our *relative, intra-attention* bias (SPA)—under one identical training protocol, data split, and evaluator. To do so we place EDM’s positional embedding (EDM Eq. (1) and Eq. (9)) on the same R1 backbone (cross-attention + dual-softmax) used by SCCM. We implement EDM’s published PE equations rather than using EDM’s full system. Any conclusion drawn here therefore bears on the *absolute-PE mechanism itself*, not on EDM as a whole system (which also has a distinct backbone and a geodesic-flow refiner).

Formulation. For each coarse-grid token at latitude/longitude (φ, λ) we form the 3D unit ray (EDM Eq. (1))

$$\mathbf{r}(\varphi, \lambda) = (\sin \lambda \cos \varphi, \sin \varphi, \cos \lambda \cos \varphi), \quad (3)$$

the same unit ray $\mathbf{r}(\varphi, \lambda)$ used by our TPB (Sec. 4.1, main). The embedding (EDM Eq. (9)) is

$$\boldsymbol{\chi} = \cos(\mathbf{W}_{\text{PE}} \mathbf{r} + \mathbf{b}), \quad \mathbf{W}_{\text{PE}} \in \mathbb{R}^{D \times 3}, \mathbf{b} \in \mathbb{R}^D, \quad (4)$$

with $\mathbf{W}_{\text{PE}}$ a learned 1×1 projection ($3 \rightarrow D$), $\mathbf{b}$ a learned bias, and $\cos$ element-wise. $\boldsymbol{\chi}$ is added to *both* token sets *before* the attention cascade ($\mathbf{x} \leftarrow \mathbf{x} + \boldsymbol{\chi}$, $\mathbf{y} \leftarrow \mathbf{y} + \boldsymbol{\chi}$). It is purely absolute (per-token, independent of the query-key pair): there is no relative-phase mechanism, which is precisely the property that distinguishes it from SPA.

Initialization. We initialize $\mathbf{W}_{\text{PE}} = \mathbf{0}$ and $\mathbf{b} = \frac{\pi}{2}$, giving $\boldsymbol{\chi} = \mathbf{0}$ at initialization.

Reproducibility. We implement only the EDM absolute sphere-coordinate embedding (unit ray Eq. (1) $\rightarrow \cos$ of a learned $3 \rightarrow D$ projection Eq. (9) $\rightarrow$ added to both token grids before the R1 coarse matcher); all other EDM components are not used.

K AAC/LAC Numerical Sanity Checks

Clamp sensitivity. The numerical safety clamp $\epsilon = 10^{-3}$ in Eq. (5) (AAC) of the main paper is active only in an extremely thin numerical pole band ($|\varphi| > 89.94^\circ$, $\sim 0.06\%$ of latitudes); all remaining pixels see the exact unclamped $\cos \varphi$.

Table 6: Sensitivity to the numerical safety clamp ϵ used in Eq. (5) of the main paper, on the full Matterport3D *test* split (15,682 pairs) with the full SCCM model. Varying ϵ over $[10^{-4}, 10^{-2}]$ leaves all metrics unchanged at reported precision, confirming that the clamp is not driving the result. PCK $\uparrow$; MAE/median in degrees $\downarrow$.

ϵ	PCK@1° $\uparrow$	PCK@3° $\uparrow$	PCK@5° $\uparrow$	MAE° $\downarrow$	Med° $\downarrow$
10^{-4}	0.275	0.665	0.806	5.36	1.88
10^{-3} (used)	0.275	0.665	0.806	5.36	1.88
10^{-2}	0.275	0.665	0.806	5.36	1.88

It acts on the non-trainable $\cos \varphi$ buffer, so the learnable α_{LAC} (entering only as $\alpha_{\text{LAC}} \log[\cdot]$) retains a well-defined gradient dominated by the unclamped majority. Tab. 6 shows that varying ϵ over $[10^{-4}, 10^{-2}]$ leaves every metric unchanged.

LAC strength. The learned LAC strengths remain near the analytic area value ($\alpha_{\text{LAC}}^A = 0.978$, $\alpha_{\text{LAC}}^B = 1.028$; init $\alpha = 1$). Overriding them to $\alpha = 1$ at inference time leaves the metrics within 10^{-4} (Tab. 7), indicating low sensitivity to the final scalar value at inference.

Table 7: LAC strength sanity check on the full Matterport3D test split. The trained SCCM checkpoint is evaluated with its *learned* per-view LAC strengths ($\alpha_{\text{LAC}}^A, \alpha_{\text{LAC}}^B$) and with α *fixed to 1 at inference time*. The two settings yield metrics within 10^{-4} of each other.

Model	$\alpha_{\text{LAC}}^A / \alpha_{\text{LAC}}^B$	PCK@1° $\uparrow$	PCK@3° $\uparrow$	PCK@5° $\uparrow$	MAE° $\downarrow$
SCCM (learned α)	0.978/1.028	0.275	0.665	0.806	5.36
SCCM ($\alpha = 1$, infer.)	1.000/1.000	0.275	0.665	0.806	5.36

L Computational Complexity Analysis

Tab. 8 reports trainable parameters and FLOPs. SCCM adds only a parameter-free yaw-periodic RoPE, an $\approx 1.1\text{K}$ -parameter TPB MLP, and one LAC scalar per covisibility side; the TPB bias costs $\approx 8.9\text{GFLOPs}$ ($\approx 1.2\%$) when recomputed every forward and nothing when cached. Tab. 9 reports latency: uncached TPB recomputation is memory-bound, but caching the content-independent TPB table reduces the deployed overhead to $\sim 5\%$. We detail the FLOPs bound and latency protocol below.

L.1 FLOPs

The chart-naïve baseline forward pass measures **751.15** GFLOPs at the medium ERP resolution 448×896 (**thop**, RTX 4090). The SCCM additions are bounded

Table 8: Computational footprint relative to the R1 chart-naïve baseline: trainable-parameter and FLOP additions at the medium ERP resolution (448×896). Inference latency is reported separately in Tab. 9.

Configuration	Trainable params	GFLOPs
(R1) chart-naïve (cross-attn+DS)	137.17 M	751.15
+ SPA (RoPE + TPB)	$+ \approx 1.1 \text{ k}$	≈ 9.0 (≈ 0.1 cached)
+ AAC (LAC)	+2	≈ 0.0
SCCM total	137.17 M	≈ 760.2 (≈ 751.3 cached)
Increase vs. baseline	$< 0.002\%$	$\approx 1.2\%$ ($\approx 0.01\%$ cached)

analytically. RoPE, a per-channel rotation of queries and keys in each attention layer, costs about 0.1 GFLOPs in total, and AAC is an N -element pointwise term with negligible cost. TPB evaluates its $32 \rightarrow 32 \rightarrow 1$ MLP on all $N^2 = 2048^2$ token pairs, ≈ 4.4 GMACs ≈ 8.9 GFLOPs, i.e. $\approx 1.2\%$ of the 751 GFLOPs baseline when the bias is recomputed every forward, and 0 when the content-independent bias is cached (Sec. L.2). The wall-clock overhead of the uncached path (Tab. 9) exceeds this compute share because it is memory-bound (an $N \times N \times 32$ intermediate) with kernel-launch overhead; caching eliminates it, leaving a deployed $\sim 5\%$.

L.2 Inference Latency Protocol

The latency values in Tab. 9 are wall-clock times on an RTX 4090, single image pair at the medium ERP resolution, mean over 100 forward passes after a 10-iteration warm-up. The pairwise tangent log-map buffer (Sec. 4.1 (ii), main) is precomputed and excluded from the per-forward timing, consistent with deployment.

TPB bias caching. The TPB bias is a fixed grid-geometry prior at inference: it depends only on the token positions, not on image content, so it can be precomputed once and reused for every pair; the cache is behavior-preserving and disabled during training. Tab. 9 reports a back-to-back A/B on a single RTX 4090: recomputing the bias each forward costs +20.9%, whereas caching it reduces SCCM’s overhead to +5.0%. We use the cached configuration for deployment.

Table 9: TPB bias caching latency. Wall-clock latency on a single RTX 4090 at medium ERP resolution, averaged over 100 timed forwards after warm-up. The cached variant precomputes the content-independent TPB bias for the fixed ERP grid and reuses it at inference time.

Method	Latency (ms)	Overhead
chart-naïve	115.76	—
SCCM (TPB uncached)	139.95	+20.90%
SCCM (TPB cached)	121.60	+5.04%

M Module Contributions and Interaction

Fig. 2 summarizes the marginal changes along the $R1 \rightarrow R2a \rightarrow R2b \rightarrow R3$ ablation in Tab. 2 (main). Yaw-periodic RoPE provides the largest $PCK@1^\circ$ gain (+3.7pp), with a $+0.19^\circ$ change in MAE. TPB reduces MAE by 0.29° with a -0.1 pp change in $PCK@1^\circ$. Adding LAC to full SPA improves $PCK@1^\circ$ by 0.9pp and reduces MAE by 0.22° . These are cumulative marginal effects; the factorial analysis below separates the $TPB \times LAC$ interaction.

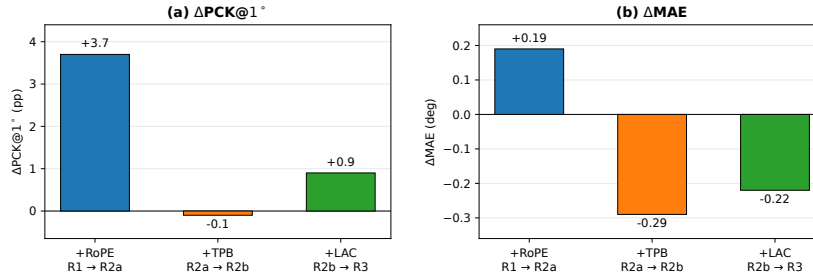


Fig. 2: Marginal changes along $R1 \rightarrow R2a \rightarrow R2b \rightarrow R3$ from Tab. 2 (main): blue +yaw-periodic RoPE ($R2a - R1$), orange +TPB ($R2b - R2a$), and green +LAC ($R3 - R2b$). (a) $\Delta PCK@1^\circ$ (pp); (b) ΔMAE (deg). Positive ΔPCK and negative ΔMAE indicate improvement.

TPB error-tail quantiles. To verify that TPB acts on the angular-error tail rather than strict threshold precision, we compare pixel-level angular-error quantiles of R2a and R2b on the full Matterport3D test set (Tab. 10). TPB leaves $PCK@1^\circ$ and the median (P50) essentially unchanged, but reduces the high-error quantiles (P90/P95/P99 by $0.41^\circ/1.80^\circ/4.96^\circ$). This confirms the interpretation in Tab. 2 (main) that TPB calibrates the pairwise metric and suppresses large-error tails rather than improving near-threshold precision.

Table 10: TPB error-tail quantiles on Matterport3D test (full split, $\sim 3.0B$ valid pixels). R2a is yaw-periodic RoPE only; R2b adds TPB (full SPA). Angular-error quantiles P_k are in degrees; in the Δ row, $PCK@1^\circ$ is in percentage points and all other columns in degrees. Differences are computed from the displayed values. TPB leaves strict precision ($PCK@1^\circ$) and the median (P50) nearly unchanged but lowers the high-error quantiles, confirming its role as tail calibration.

Row	$PCK@1^\circ \uparrow$	$MAE \downarrow$	$P50 \downarrow$	$P75 \downarrow$	$P90 \downarrow$	$P95 \downarrow$	$P99 \downarrow$
R2a (RoPE)	0.267	5.87	1.93	4.18	9.29	19.49	94.23
R2b (full SPA)	0.266	5.58	1.95	4.13	8.88	17.69	89.27
Δ	-0.1	-0.29	+0.02	-0.05	-0.41	-1.80	-4.96

Factorial decomposition: the two priors interact. The cumulative ladder of Tab. 2 (main) credits each gain to the last added term. To separate the roles of TPB and LAC we train the missing combination (yaw-periodic RoPE + LAC, no TPB) under the identical protocol and evaluate all four combinations, completing a 2×2 factorial over the R2a base (Tab. 11). Individually, TPB and LAC each change PCK@1° by -0.1 pp; together they improve it by $+0.8$ pp. The resulting interaction is $+1.0$ pp. Both priors reduce MAE individually (-0.29° and -0.34°), with conditional reductions of -0.17° and -0.22° , respectively. The combination improves strict PCK@1°, consistent with their complementary roles: LAC adjusts the matching mass retained by the area-aware gate, while TPB adjusts its distribution under the spherical metric. This distribution determines the expected target coordinate (Sec. I).

Table 11: TPB×LAC factorial on Matterport3D test (15,682 pairs): all four combinations over the R2a (yaw-periodic RoPE) base under the shared protocol of Tab. 1. Effects and the interaction are computed from the displayed values.

Cell	TPB	LAC	PCK@1°↑	PCK@3°↑	PCK@5°↑	MAE↓
R2a (RoPE)	✗	✗	0.267	0.653	0.796	5.87
R2b (+TPB)	✓	✗	0.266	0.652	0.799	5.58
+LAC (no TPB)	✗	✓	0.266	0.658	0.802	5.53
R3 (SCCM)	✓	✓	0.275	0.665	0.806	5.36
Effect	Δ PCK@1° (pp)			Δ MAE (deg)		
TPB alone (R2b–R2a)	-0.1			-0.29		
LAC alone	-0.1			-0.34		
TPB given LAC	$+0.9$			-0.17		
LAC given TPB (R3–R2b)	$+0.9$			-0.22		
Interaction	$+1.0$			$+0.12$		

N Analysis of Remaining Errors

SCCM’s residual errors fall into two groups. **(a) Geometry-specific failures** occur outside the intended upright-ERP setting: a large camera pitch/roll breaks the gravity-aligned prior and accuracy drops sharply (PCK@1° $0.273 \rightarrow 0.034$ at a 30° tilt; Fig. 3, Sec. F). Within upright ERP, the polar safety clamps are inactive because no coarse-grid token lies in their pole band (Secs. A and K); the separate handling of exact antipodal token pairs is described in Sec. A. **(b) Generic dense-matching failures** remain in texture-less, reflective/HDR, low-overlap, and depth-discontinuity regions; these are shared by perspective and ERP matchers alike (cf. RoMa [5], EDM [6], LoFTR [12]), are driven by visual ambiguity rather than chart geometry, and are orthogonal to the spherical

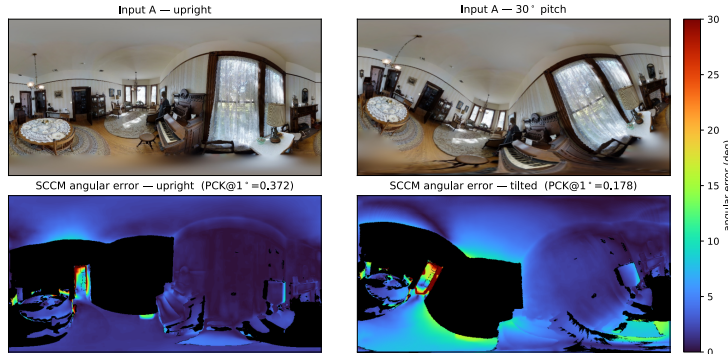


Fig. 3: SCCM’s off-gravity failure mode, shown on a single Matterport3D test pair (the *same* pair throughout). *Top*: input view A upright vs. under a synthetic 30° off-gravity pitch (Sec. F), which tips the ERP chart off the gravity horizon. *Bottom*: SCCM’s per-pixel angular error for the warp $A \rightarrow B$ (black = non-covisible; colorbar 0° to $\geq 30^\circ$). Upright, SCCM attains $\text{PCK}@1^\circ = 0.37$ on this pair (mostly low error); under tilt the gravity-aligned prior no longer matches the tilted chart and accuracy drops markedly ($\text{PCK}@1^\circ = 0.18$), with error concentrated where the chart distortion is largest. Holding the pair fixed isolates tilt—not scene content—as the cause; full-set tilt averages are in Tab. 5. This is a geometry-specific failure outside the intended upright-ERP setting.

priors—requiring upstream changes (a stronger encoder, photometric-invariant features, occlusion handling).

N.1 Certainty-gated vs. ungated reconstruction

The main reconstruction protocol uses a shared certainty gate $\tau = 0.5$ for all methods. Without this gate, SCCM’s mean accuracy is affected by a rare low-certainty tail, while its median accuracy and majority-pair performance remain better than chart-naïve.

O Refinement-Stage Diagnosis

To locate where the mechanism-isolated SCCM gain enters the pipeline, we measure $\text{PCK}@1^\circ$ after the coarse stage and after each refinement stage of the shared cascade, on the full Matterport3D test split ($\sim 3.0\text{B}$ valid pixels). Tab. 12 shows that SCCM’s advantage over the chart-naïve scaffold is already present at the coarse-anchor stage (+4.1 pp) and remains essentially unchanged through refinement (+4.5 pp at the final dense output), with small non-monotonic fluctuation across intermediate stages (+4.1 to +5.4 pp). The architecturally unchanged RoMa V1 refiner provides a similar coarse-to-final lift for both models (+10.8 pp for chart-naïve, +11.2 pp for SCCM), preserving and slightly amplifying the coarse-stage advantage rather than creating it. This supports our interpretation

that the ERP-specific gain enters primarily through coarse-anchor formation rather than being created by the refiner.

Table 12: Refinement-stage PCK@1° diagnostic on the full Matterport3D test split. PCK@1° after the coarse stage and after each refinement stage. SCCM’s advantage is present before refinement and preserved through the architecturally unchanged RoMa V1 refiner; the similar coarse-to-final lift for both models indicates that the gain enters primarily through coarse-anchor formation rather than being created by the refiner.

Stage	chart-naïve	SCCM	Δ (pp)
coarse anchor (1/16)	0.122	0.163	+4.1
refiner stage (1/8)	0.133	0.187	+5.4
refiner stage (1/4)	0.188	0.234	+4.6
refiner stage (1/2)	0.219	0.263	+4.4
final dense	0.230	0.275	+4.5

P Outdoor Evaluation on Holo360D

This section details the outdoor experiment summarized in Tab. 1(c) of the main paper. Holo360D [10] is an in-the-wild dataset captured with a handheld LiDAR scanner coupled to a 360° camera along continuous trajectories. Its outdoor sequences are far deeper and sparser in supervision than Matterport3D (median scene depth 4.3 m vs. 2.0 m; 54% vs. 95% of ERP pixels carry valid depth). The median relative tilt between paired views is 12°, with 79% of pairs exceeding 5°.

Protocol. The three retrained matchers of Tab. 1 (main)—the ERP-retrained RoMa V1 (V1-GP), the chart-naïve scaffold (R1), and SCCM—are initialized from their Matterport3D checkpoints and trained on Holo360D with a shared training budget (25k optimizer steps, learning-rate scale 0.1, same augmentation and loss), on scene-disjoint splits of 5 training, 3 validation, and 4 test scenes, and score 8,000 held-out test pairs (2,000 per test scene; overlap 0.3–0.8) with the angular evaluator of the main paper. For SCCM’s Holo360D training, the RoPE rotation angles are multiplied by $\gamma \sim U[0, 1]$ at each forward pass; evaluation uses $\gamma=1$. This stochastic regularization is specific to Holo360D training.

Result. SCCM leads on all seven metrics (Tab. 13), including the two stricter thresholds not shown in the main table (+1.8 pp at 0.35°, with a +2.6 pp margin over R1 at 1°).

Zero-shot transfer. Applying the same Holo360D-trained models *unchanged* to a second outdoor corpus (Mapillary Metropolis [8]; 2,484 street-level pairs resampled to match the rotation distribution of the Holo360D test set) tests transfer

Table 13: Outdoor training on Holo360D (8,000 test pairs). All three matchers start from their Matterport3D checkpoints and are trained with the settings above; SCCM leads on every metric, including the two error statistics. Best in **bold**.

Method	PCK@0.35°↑	PCK@0.5°↑	PCK@1°↑	PCK@3°↑	PCK@5°↑	MAE°↓	Med.°↓
RoMa V1 (ERP-retrained)	0.123	0.182	0.322	0.592	0.726	5.66	2.12
chart-naïve (R1)	0.119	0.180	0.331	0.626	0.761	5.02	1.93
SCCM	0.137	0.202	0.357	0.645	0.776	4.94	1.76

without any adaptation. SCCM retains its lead at tight thresholds (+1.9 pp PCK@0.35°, +0.5 pp PCK@1° over R1) but not at 3°–5° or in mean angular error (10.08° vs. 9.39° for R1), i.e. its zero-shot advantage is confined to precision.

Q Additional Zero-Shot Baselines on ERP

Tab. 14 extends Tab. 1 of the main paper with four further published matchers, evaluated zero-shot on ERP under the same angular evaluator, each released checkpoint run at its own operating resolution with the ground truth on that grid (as for EDM in Sec. R). All four fall well below EDM, the strongest released ERP-native baseline, indicating limited zero-shot ERP performance of these checkpoints.

Table 14: Additional zero-shot baselines (PCK@1°). [†]Sparse matchers scored on their own matches only. EDM from Tab. 1 for reference.

Method	Matterport3D	Stanford2D3D
DKM [3] (CVPR’23)	0.035	0.026
MASt3R [†] [7] (ECCV’24)	0.057	0.028
LoFTR [†] [12] (CVPR’21)	0.105	0.058
VGGT [13] (CVPR’25)	0.000	0.000
EDM [6] (reference)	0.163	0.104

R Additional Controls: Seeds, Resolution, and Pose Solver

Training-seed variance. The main ablation trains each configuration once. To bound training-side variance we retrained the two endpoints of the ablation, chart-naïve (R1) and SCCM, with two additional seeds each under the identical protocol and re-scored the full Matterport3D test split (Tab. 15). PCK@1° ranges from 0.272 to 0.283 for SCCM and from 0.223 to 0.230 for R1. The per-configuration sample standard deviation is 0.4–0.6 pp, whereas the +0.9 pp LAC

step of Tab. 2 is only about twice this scale and, being measured from single runs, should be read as indicative on the training side.

Table 15: Multi-seed retraining of the ablation endpoints (Matterport3D test, PCK@1°). Results from three training runs per model; run 1 is the paper result. Standard deviations are computed within each model.

Configuration	run 1 (paper)	run 2	run 3	s.d. (pp)
chart-naïve (R1)	0.230	0.230	0.223	0.4
SCCM	0.275	0.283	0.272	0.6

EDM resolution control. EDM [6] is evaluated at its native training resolution (320×640); its ground truth and metrics are computed on that grid. Because no re-scaling of SCCM to EDM’s resolution is canonical, we bracket it: (i) down-sampling SCCM’s input to 320×640 and back (matching input information only) gives PCK@1° 0.273; (ii) matching input, ground truth, *and* scored pixels to EDM’s grid (the same 1.47B evaluated pixels) gives 0.279. Both straddle the reported 0.275: re-scaling moves SCCM by $-0.2/+0.4$ pp while its gap over EDM (0.163) stays at 11.0–11.6 pp.

Pose solver. We compare the shared ray-space 8-point+RANSAC estimator with Robust 360-8PA [11] on fixed matches and inliers. The common ≥ 200 -confident-matches filter over the four methods yields 15,112 pairs (Tab. 16). AUC@5° changes by at most 0.01 pp, with the ranking unchanged.

Table 16: Solver swap on 15,112 Matterport3D test pairs: Pose AUC@5° with our 8-point+RANSAC estimator vs. Robust 360-8PA on identical matches and inliers.

Method	8-pt+RANSAC	Robust 360-8PA
EDM [6]	8.28	8.29
RoMa V1 (ERP-retrained)	11.69	11.68
chart-naïve (R1)	14.17	14.16
SCCM	17.96	17.96

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